

KIMCHAEK SONGJIN, Korea, Democratic People's Republic of

WMO: 470250

Lat: 40.667N

Lon: 129.200E

Elev: 23

StdP: 101.05

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-10.3	-25.2	0.4	-9.4	-22.2	0.5	-7.8	8.0	-5.4	6.8	-3.8	1.6	320	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	28.7	23.9	27.1	23.2	25.8	22.5	25.1	27.6	24.2	26.3	23.4	25.2	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.3	26.9	23.5	18.4	25.9	22.7	17.5	24.9	77.3	27.7	73.5	26.4	70.2	25.3	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.6	5.2	-15.5	32.2	1.9	1.5	-16.8	33.3	-17.9	34.2	-19.0	35.1	-20.3	36.2		
(5)			-16.5	26.6	1.7	1.5	-17.7	27.6	-18.8	28.5	-19.7	29.3	-21.0	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.4	-2.0	2.8	8.6	12.9	17.1	21.2	22.9	18.7	12.8	5.4	-0.9
	DBStd	9.42	3.23	3.32	3.36	3.25	3.20	2.47	2.83	2.09	2.32	2.95	3.89	3.65
	HDD10.0	1538	416	336	224	64	8	0	0	0	0	9	143	337
	HDD18.3	3422	674	569	481	291	170	54	6	1	22	171	388	596
	CDD10.0	1448	0	0	1	23	99	212	348	401	261	97	6	0
	CDD18.3	291	0	0	0	0	3	16	96	143	33	1	0	0
	CDH23.3	1186	0	0	0	3	21	46	371	652	93	2	0	0
	CDH26.7	193	0	0	0	0	3	8	72	100	11	0	0	0
Wind	WSAvg	1.5	1.4	1.6	1.8	2.0	1.7	1.3	1.2	1.4	1.5	1.6	1.5	1.4
Precipitation	PrecAvg	704	23	17	18	36	59	71	155	136	90	41	39	26
	PrecMax	1240	90	60	86	91	130	222	394	356	288	140	105	75
	PrecMin	480	5	1	0	5	5	11	4	16	16	1	1	5
	PrecStd	180	17	15	20	23	32	53	103	80	73	31	28	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	8.7	16.2	22.5	26.0	27.6	31.0	31.2	28.2	22.7	17.5	10.0
		MCWB	3.4	4.2	9.3	13.3	15.9	19.0	24.6	25.8	21.3	16.6	11.8	6.4
	2%	DB	4.3	6.1	12.0	18.7	22.7	24.4	28.6	29.0	25.6	20.7	15.1	7.8
		MCWB	1.3	3.2	7.3	10.8	14.7	18.3	24.1	24.8	20.5	16.0	11.0	5.1
	5%	DB	2.9	4.6	9.6	16.1	20.1	22.1	26.7	27.5	24.1	19.4	13.3	6.2
		MCWB	0.4	2.2	5.8	9.8	13.8	18.4	23.6	24.0	19.8	15.1	9.8	3.5
	10%	DB	1.7	3.1	7.9	14.0	17.7	20.7	25.3	26.3	22.9	18.1	11.5	4.6
		MCWB	-0.5	1.1	4.8	8.9	13.0	18.0	22.8	23.4	19.2	14.5	8.4	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	5.4	10.0	14.0	17.5	20.8	26.3	26.7	23.3	18.9	13.2	7.3
		MCDB	5.9	7.4	14.8	21.1	22.6	23.2	29.6	30.0	26.2	20.5	15.6	9.0
	2%	WB	2.2	3.8	7.7	12.1	15.7	20.0	24.9	25.6	21.8	17.1	11.7	5.6
		MCDB	3.7	5.5	11.2	16.8	20.4	22.2	27.4	28.2	23.9	19.5	14.6	7.5
	5%	WB	0.9	2.5	6.4	10.6	14.8	19.2	23.9	24.7	20.8	16.1	10.2	4.0
		MCDB	2.3	4.2	9.1	14.7	18.3	21.3	26.1	26.8	22.9	18.6	12.7	5.7
	10%	WB	-0.2	1.4	5.2	9.5	13.9	18.4	23.1	23.9	20.0	15.0	8.8	2.5
		MCDB	1.3	3.0	7.4	13.0	17.0	20.3	25.1	25.8	22.1	17.4	11.2	4.2
Mean Daily Temperature Range	5% DB	MDBR	8.0	7.5	7.2	8.0	6.7	4.7	4.2	5.2	7.2	8.5	7.9	7.9
		MCDBR	8.8	8.8	10.2	12.4	12.0	8.5	7.3	7.2	8.7	9.7	10.2	8.8
		MCWBR	7.2	6.9	6.7	6.6	6.1	4.4	4.2	4.1	5.4	6.5	7.4	7.1
	5% WB	MCDBR	7.9	8.3	9.3	10.7	9.4	6.0	6.3	5.8	6.9	8.3	9.3	8.2
		MCWBR	6.9	6.7	6.7	6.4	5.6	3.8	4.1	3.8	5.0	6.2	7.1	7.1
Clear-Sky Solar Irradiance	taub	0.357	0.409	0.481	0.551	0.582	0.605	0.561	0.508	0.450	0.444	0.402	0.360	
	taud	2.123	2.009	1.869	1.768	1.772	1.798	1.983	2.121	2.203	2.081	2.100	2.144	
	Ebn at Noon	764	774	757	735	724	708	736	765	785	730	702	724	
	Edh at Noon	108	141	182	215	219	214	176	150	129	130	111	98	
All-Sky Solar Radiation	RadAvg	2.22	3.18	4.41	5.12	5.28	4.99	4.50	4.53	4.20	3.38	2.27	1.94	
	RadStd	0.16	0.25	0.28	0.32	0.53	0.51	0.56	0.35	0.41	0.27	0.35	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page